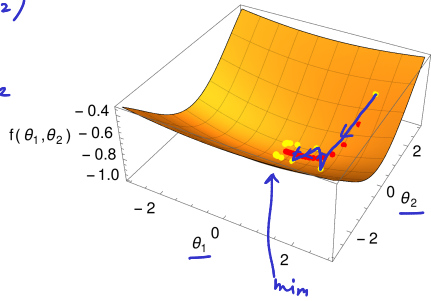
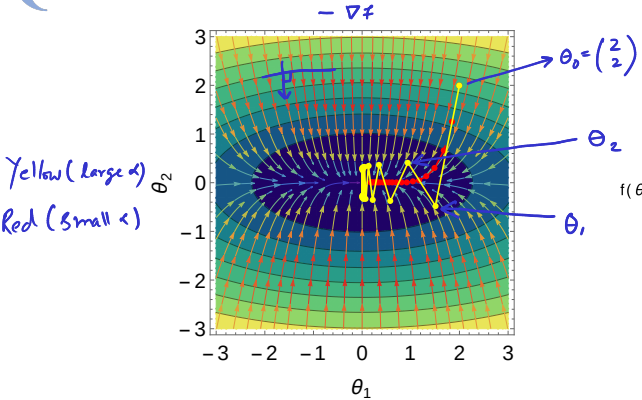


"Gradient descent" for local minimum f

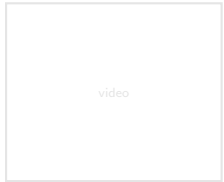


The function **decreases fastest** in the direction **opposite** to the gradient:

$$\theta_{i+1} = \theta_i - \alpha \nabla f(\theta_i) \quad \text{amount of movement} = |\alpha| |\nabla f|$$

Step size is $\alpha > 0$, $\theta = (\theta_1, \theta_2)$, and i represents i th iteration.

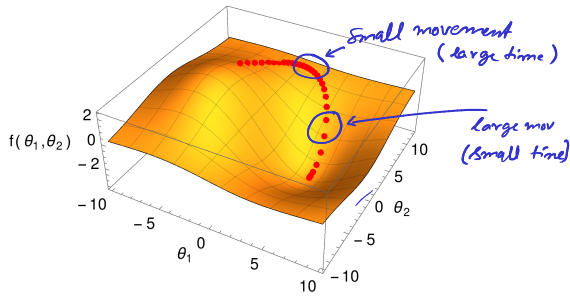
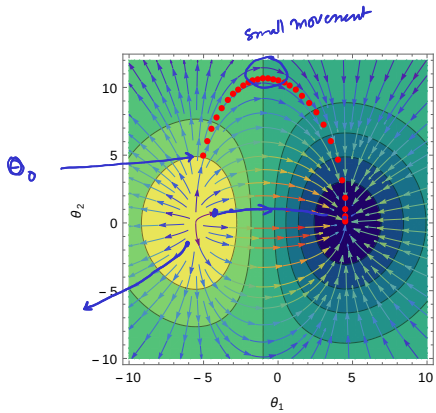
$$f(\theta_0) \geq f(\theta_1) \geq f(\theta_2) \cdots f(\theta_n) \approx \min f$$



Gradient descent for local minimum

amount of movement = $\alpha ||\nabla f||$

$$\theta_{i+1} = \theta_i - \alpha \nabla f$$



$$f(\theta_1, \theta_2) = -(1 + \theta_1) \exp\left(-\frac{(\theta_1)^2 + (\theta_2)^2}{50}\right)$$

Initial point $\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} -5 \\ 5 \end{pmatrix}$

